SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME: Cryptography and Network Security**

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO* (BL5, BL6)

Assessment Tool Weightage: 10%

QUESTIONS: 20 **TOTAL MARKS: 20**

Annexure A
MOCK INTERVIEW

Code of Conduct:

I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: E. Suchier

Annexure A

MOCK INTERVIEW

1. **Analyze the weaknesses of SSL and explain why TLS is considered more secure in modern web systems.**

   SSL has older security weaknesses and vulnerable protocols. TLS is more secure because it uses stronger cryptographic algorithms and improved handshake mechanisms.

2. **Apply the TLS handshake process to explain how secure session keys are established between a client and server.**

   The client and server exchange information and authenticate using a digital certificate. They then perform key exchange to establish a shared session key.

3. **Compare SSL and TLS in terms of encryption mechanisms and identify key improvements in TLS.**

   TLS provides stronger encryption, authentication, and integrity than SSL. TLS 1.3 also removes outdated algorithms and improves handshake security.

4. **Evaluate a scenario where improper certificate validation in TLS leads to a security breach. What could go wrong?**

   Improper certificate validation can allow a Man-in-the-Middle attack. The attacker can intercept or modify sensitive information.

5. **Analyze the role of digital certificates in SSL/TLS and explain how they ensure authentication.**

   A digital certificate binds a server's identity to its public key and is issued by a trusted CA. It helps the client authenticate the server.

6. **Apply your understanding of SET protocol to explain how it protects credit card transactions in e-commerce.**

   SET protects online credit-card transactions using encryption and digital certificates. It provides confidentiality, authentication, integrity, and non-repudiation.

7. **Evaluate the limitations of the SET protocol in current online payment systems.**

   SET is complex, expensive to deploy, and requires certificate management. Therefore, it is rarely used in modern payment systems.

8. **Compare SET protocol with TLS-based payment security and justify which is more practical today.**

   SET provides detailed payment security but is complex, while TLS is simpler and widely supported. **TLS-based payment security is more practical today.**

9. **Analyze a malware attack scenario and differentiate whether it is caused by a virus, worm, or Trojan.**

   A virus attaches to files, a worm spreads automatically across networks, and a Trojan disguises itself as legitimate software.

10. **Apply appropriate mitigation techniques for a worm attack spreading across a corporate network.**

    Isolate infected systems, apply security patches, use firewalls and IDS/IPS, and update antivirus software to stop further spread.

11. **Evaluate the impact of Trojan attacks on system confidentiality and integrity with an example scenario.**

    Trojans can steal confidential information and modify system data. For example, a fake application may secretly capture passwords.

12. **Compare viruses, worms, and Trojans based on propagation method and system impact.**

    Viruses require infected files, worms self-replicate through networks, and Trojans rely on user deception. All can damage or compromise systems.

13. **Analyze the working of a packet-filtering firewall and identify its limitations in modern networks.**

    It checks packet details such as IP addresses, ports, and protocols against rules. Its limitation is that it has limited application-level inspection.

14. **Apply firewall rules to block unauthorized access while allowing legitimate traffic in a given scenario.**

    Allow required services such as HTTPS on port 443 and restrict administrative ports such as SSH. Deny all unnecessary incoming traffic.

15. **Evaluate the effectiveness of stateful inspection firewalls over stateless firewalls.**

    Stateful firewalls track active connections and make more intelligent decisions. They are generally more effective than stateless firewalls.

16. **Compare different types of firewalls and recommend the most suitable one for an enterprise network.**

    Packet-filtering, stateful, proxy, and next-generation firewalls are common types. NGFWs are suitable for enterprises because they provide advanced traffic inspection and security features.

17. **Analyze how an IDS detects malicious activities using signature-based and anomaly-based techniques.**

    Signature-based IDS detects known attack patterns, while anomaly-based IDS detects deviations from normal behavior. Combining both improves detection.

18. **Apply IDS/IPS mechanisms to prevent a distributed denial-of-service (DDoS) attack.**

    IDS detects abnormal traffic and generates alerts, while IPS can block or rate-limit malicious traffic. Firewalls and traffic filtering can provide additional protection.

19. **Evaluate the differences between IDS and IPS in terms of response to detected threats.**

    IDS detects suspicious activity and alerts administrators, whereas IPS detects and automatically blocks malicious traffic.

20. **Propose improvements to reduce false positives in an IDS while maintaining high detection accuracy.**

    Tune detection rules, establish accurate traffic baselines, update signatures, and adjust alert thresholds. Regular monitoring helps maintain detection accuracy.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides	Provides reasonable analysis with	Superficial or partially structured responses	No analytical thinking demonstrated

		well-structured, logical answers	some structure		
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism